

Installation & Application Instructions for OSP MicroCore® Fibre Optic Cable



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The following document provides generic technical requirements and characteristics for MicroCore® installation procedures and is designed to assist installation personnel and contractors to understand the unique characteristics of working with AFL Telecommunications (AFL) MicroCore® cable. Because this product is designed to be blown instead of pulled, there are several performance differences from standard outside plant duct and cable specifications. In all instances, the installation equipment, material, and AFLs installation procedures shall take precedent over the general instructions provided in this document.

1.0 GENERAL INFORMATION

The instructions in this section explain, in general, how to install AFL MicroCore® cable into microducts. Because MicroCore® is designed by AFL specifically to be blown, it incorporates several performance differences from standard outside plant cable and special procedures and handling instructions must be applied for this product.

2.0 SAFETY

Personal Protection Equipment AFL strongly recommends the use of approved personal protection equipment for outside plant construction in the performance of this procedure. Proper eye and ear protection is especially important when working around compressed air. All protection equipment must meet the relevant safety requirements.

3.0 CABLE HANDLING

3.1 REEL TRANSPORTATION / HANDLING

- 3.1.1 While loading or unloading cable reels, care must be taken to prevent collision with anything that may damage the cable reel.
- 3.1.2 The reel should not be rolled a long distance. If rolling is necessary, it should be rolled for only very short distances and in the direction indicated by the arrow shown on the reel.
- 3.1.3 The reel should be located on a flat surface and blocks/chock placed to prevent it from rolling in either direction.
- 3.1.4 The cable on the reel should be covered until just prior to installation to protect the jacket from exposure to the sun.
- 3.1.5 Once the protective covering is removed, inspect the reel for imperfections including nails and broken flanges which may cause problems during the installation process.
- 3.1.6 Release the inner end of the cable which is connected to the outside of the reel.
- 3.1.7 For details on reel handling, refer to the separate document 'Cable Storage and Handling Guide'.

3.2 PRE-INSTALLATION CABLE OTDR TESTING

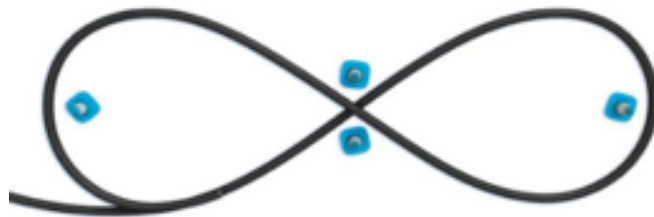
Contractor may elect to test the cable prior to installation to protect the contractor from taking responsibility of damage to the cable during shipment. It is strongly recommended that the cable be tested using an OTDR and the values recorded prior to installation at 1310nm and 1550nm to determine dB loss per kilometer, length of fibre, continuity of all fibres, and any fibre abnormalities

3.3 MINIMUM BEND RADIUS

Care must be always taken to ensure the cable is not bent more than Minimum Bend Radius - Under Load, that is 20 times the outside cable diameter, as specified in relevant cable technical data sheet.

3.4 KINKING

Care must be always taken to ensure the cable is never kinked. If some slack is pulled from the reel to the installation equipment, monitor any loops that form as the slack is taken back up. Loops of slack that are pulled straight can easily kink the cable. During excess slack at pulling locations, it is recommended to lay the cable in a figure "8" to prevent cable twist. Kinking fibre optic cable of any type will damage internal cable components, including the glass fibres within.



3.5 CRUSHING

MicroCore® cable has reduced crush resistance versus typical OSP Loose Tube fibre cable. Care must be always taken to insure the MicroCore® cable is never crushed, both on and off the reel. This is always an issue as the MicroCore® cable can be crushed while on the reel or on the ground. MicroCore® cable on the ground in a figure 8 must be always protected from vehicular as well as pedestrian traffic.

3.6 TENSION

MicroCore® cable is specifically designed by the manufacturer to be installed by blowing technology. If a condition occurs when cable pulling is required, contractor is required to set a breakaway tension of 85% of Maximum Pulling Tension - Install, as specified in cable technical data sheet.



4.0 MICROCORE® APPLICATION

MicroCore® optical cables provide a unique blown solution to crowded underground ducts where conventional cable designs are not an option. MicroCore® can also be used for 'overriding' existing networks and conduits, which might otherwise be deemed as 'at capacity'. MicroCore® cables are jetted or blown through a network of microducts using compressed air. Conduit systems can be laid and microducts blown in as needed, giving installers the flexibility of deploying fibre only when required, thereby reducing the initial investment costs.

Whether the need is for high fibre density or small cable diameters, AFL MicroCore® has the solution ideal for short and long-haul, point-to-point, point-to-multipoint backbone applications. Designs are focused on minimal cable and duct diameters to ensure a cost-effective installation solution. The flexible, lightweight cable design incorporating a low friction jacket allows it to be blown over existing cables within a duct, thereby expanding the capacity of that duct which would have otherwise been deemed full.

5.0 CABLE PREPARATION

Safety measures must be taken before commencing the cable / fibre stripping procedure. Ensure suitable protective personal equipment and follow local relevant safety procedures. Appropriate stripping tools must be used for easy access through the sheath layers. Try not to damage the fibres during this process.

Refer relevant AFL's Cable Stripping guide for a comprehensive step-by-step cable stripping instruction. Fibre ends must be properly cleaved to provide satisfactory visual and geometrical end-face parameters. Use fibre end inspection tools to ensure fibres and connectors are free of debris. Ensure the tools used for cable preparation are calibrated and of appropriate size to the cable.

6.0 CONSTRUCTION LAYERS

This section details multiple elements in the cable construction that must be taken into consideration during installation.

6.1 TERMITE RESISTANT BARRIERS

There are many species of termites which are economically significant in damaging buildings and infrastructure, especially in Australia, which is why a particular Polyamide (Nylon) compounds (as per AS1049) of appropriate thickness can be used against termite attack. Although termites do not derive any nutritional value from cable elements, they chew on cable when it's an obstacle in their foraging activities.

A pristine Nylon sheath will protect the cable from termite pincers as it forms a tough, shiny, and slippery barrier upon which the termite mandibles cannot get through and easily slips off. In addition to its termite resistant properties, due to its surface hardness, smooth texture, and low coefficient of friction, Nylon can be used as an outer cable sheath for cables installed through conduit with lower pulling force.



6.2 SACRIFICIAL OUTER JACKETS

The function of a sacrificial jacket is to protect the inner jacket from damage. This is quite a common practice for conventional Nylon jacketed loose tube cables which may be subjected to surface damage during the rigours of installation. If the Nylon's pristine surface is scratched, it can create a surface upon which termite mandibles can penetrate through. AFL uses a thin Polyethylene Sacrificial sheath for MicroCore® cable type that comes in blue colour as standard, but other colours are also available upon request.



6.3 UV STABILISED CABLE

Exposure to sun's UV rays, will make polymers discolour over time and go brittle. It is significant to differentiate and understand whether the cables are going to be exposed to UV rays for a long-term or short-term; hence, a UV stabilised sheath must be used as an outer most cable jacket if necessary. Cable exposure to UV only during storage and transport prior to installation is considered a short-term. AFL's Application Engineers can be a resource for providing more information regarding UV stabilisation of the cable to be purchased, based on the cable application.

7.0 CABLE INSTALLATION TECHNIQUES

Manuals and guidelines from the manufacturers of tools, equipment and accessories like lubricants used for installation must be strictly followed to prevent from any damage to the cable. Please refer to AFL MicroCore® cable tech data sheets for cable tech details. Factors that influence the installation and different techniques to install MicroCore® cables are detailed in this section. .

7.1 DUCT TYPE

HDPE, PVC and fibre reinforced glass or plastic are some of the materials used in duct manufacturing. Ducts and micro ducts should be selected based on the soil conditions, installation environment, depth of installation and fibre optic bandwidth capacity to ensure duct's stability and durability. Ensure to obtain necessary permits and adhere to local and state regulations before commencing or altering any duct installations.

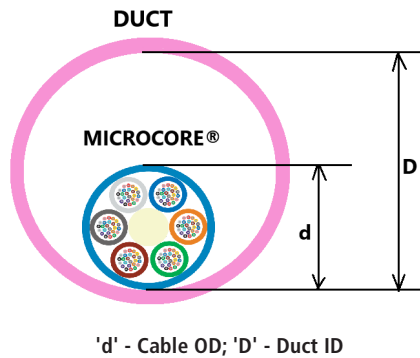
Sub-ducting can be used to provide physical or electrical separation between cables installed within the duct and any service installed within the larger duct. Ensure the Australian Standards are followed while sub-ducting cables from different services. Make sure appropriate backfilling material is used if bedding or backfilling over the duct is necessary, to not damage the ducts during installation.

Avoid any 90-degree bends in the duct. It is recommended not to install additional cables into a duct with more than two 90-degree bends due to high friction at bends between the cables and duct. Ensure sweeping bends of appropriate size are used to maintain the cable's mechanical and optical characteristics. Please refer AFL MicroCore® cable tech data sheets for tensile and minimum bend radius details.



7.2 FILL RATIO

Outer Diameter (d) and/or overall cross-sectional area of the cable with respect to the Inner Diameter (D) and/or overall cross-sectional area of the duct will determine the duct fill ratio. Fill ratio varies for any additional cable to be installed into a preoccupied duct. Consider future network expansions while planning cable installation in ducts. Follow relevant Australian standards to check the fill ratio for the cable proportional to the duct space.



7.3 HAULING (PULLING)

Hauling or pulling method is adopted when the cable length for installation is not continuous or not more than few hundred metres. It is recommended to ensure the ducts are free of any catch points that may affect the rope and/or cable access. It is preferred to use recommended lubricant on either the duct inner wall or the cable surface to facilitate maximum blowing distance. This process is to maintain the friction between elements for maximum throughput. Ensure the lubricant used is compatible to the fibre optic cable under installation to prevent the cable from any performance issues or cable jacket from degradation. Cable is pulled into the ducts either manually or using a puller machine depending on the location and length of installation using pre-installed rope in the duct.



7.4 JETTING

Cable jetting is the process of placing a fibre optic cable into an existing duct with the use of specialised installation equipment along with a high-volume air compressor. This type of installation uses a high pressure / low volume source of air to help pull the cable through the duct with the use of a Cable Carrier (A) or Parachute (B).



A



B

The installation process starts by properly attaching a fibre optic pulling grip, break-away swivel and cable carrier or parachute onto the end of the cable. The cable is then properly placed into the equipment and tractor drive mechanism. The air pressure is then applied which causes the carrier or parachute to build up pressure and form a seal. Both the speed of the drive mechanism and the amount of air pressure is then independently adjusted so that they both work together throughout the installation. The cables overall pulling force should not exceed the specified maximum tensile rating as per AFL's cable technical datasheet. The drive mechanism is designed to only assist the air flow in moving the cable forward and not to push or force the cable into the duct.

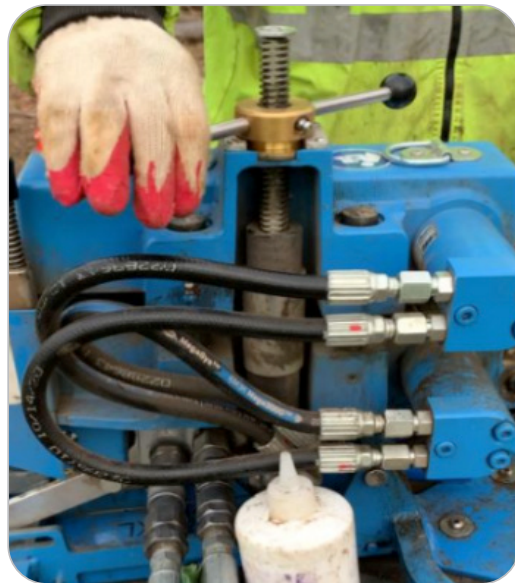
7.5 BLOWING

Blowing process use a low pressure / high-volume source of air like jetting method, but to help 'float' the cable through the duct and reduce its overall contact with the inner wall. Cable tips / end caps can also help to avoid catch points within the duct.

Determine the maximum blowing distance by analysing the route as recommended by blowing equipment suppliers. Blowing technique also referred in field as 'pushing' the cable depending on the air flow and equipment used. Piston and laminar flow are the two types of blowing unit where piston is not used in the laminar flow technique allowing complete air flow through the duct. Pushing force is higher for piston blowing unit as the resultant force has a factor of piston area multiplied to the air pressure applied.

Duct fill, cable flexibility and friction coefficient are the key factors influencing the blowing distance. It is preferred to use recommended lubricant on either the duct inner wall or the cable surface to facilitate maximum blowing distance. This process is to maintain the friction between elements for maximum throughput.

Typically, the installation process starts by properly placing the cable into the equipment and starting the tractor drive mechanism to install the cable approximately 15 to 30m into the duct without air and under low tension. This process is used to help prevent the cable from being pushed out once the initial air is applied. The air pressure is then applied in small increments and allowed to stabilize prior to starting the drive mechanism. Once the stabilization is complete, both the speed of the drive mechanism and the amount of air pressure need to be independently adjusted so that they both work together throughout the installation. Hand assist the payoff reel (if not powered) to reduce the tension on the installation machine and to prevent the cable from exceeding the max installation tension as specified in AFL's cable technical datasheet.



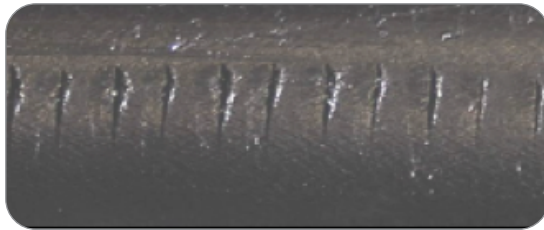
8.0 CABLE INSPECTION

Inspect the cable as often as possible and look for any unusual signs such as:

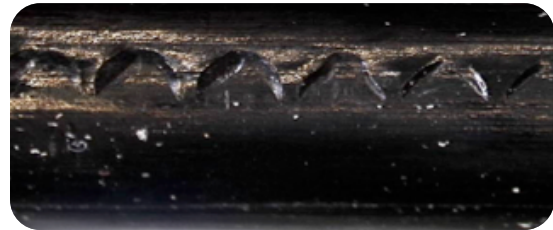
- Oval/Flat Cable
- Jacket Damage

Note: The following is information relating Sherman & Reilly Cable Jet blowing equipment:

- Light parallel line marks are expected.
- Deeper triangular marks are bad and indicate that the cable is riding on top of the Drive Train.



Good Marks - Light parallel lines
indicate side of rollers are contacting the cable



Bad Marks - Deep triangular marks
indicate cable is riding on top of the rollers

9.0 REINSTATEMENT OF SITE ON COMPLETION

Ensure that all excavation works are done to high standard with minimum disruption to the surrounding environment as much as possible. Any results that show damage or degraded characteristics of the cable under test must be rectified as per standard procedures. In any case of damage made to the property or assets in the job site, customers must be notified, and appropriate repair or alterations must be made. On completion of the work remove all plant, materials and other things that may have been brought upon the site in aid of works. Try to leave the site in a similar or better condition to how it was before the commencement of job. Ensure final sign-off on the work is received from the customer.

10.0 DOCUMENTATION

It is good practice to complete a detailed As-Built drawing and update as necessary after maintenance or repairs. Record all changes to designs and installation arrangements and forward to all stakeholders. AFL's Application Engineers can be a resource for providing information regarding previously purchased cable. Cable test reports must be part of the work completion reports. Detailed recordkeeping and documentation often are beneficial for future reference. Follow customer requirements and adhere to local, state, federal and industry codes, and regulations.